

Wei-Jin Huang

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Research Summary

I am currently a Ph.D. student at Sun Yat-sen University, specializing in Humanoid Learning and Video Action Understanding. I have a strong interest in deep learning and computer vision, with my current research focus on temporal action understanding. I have a deep passion for all computer science things, and I aim to make significant contributions to the field through extensive and meaningful research.

Education

Ph.D., Computer Science and Technology - Sun Yat-sen University (SYSU), Guangzhou, China

2024.09 - Present

- Specialization: Humanoid Learning, Video Action Understanding

B.E., Network Engineering - South China University of Technology (SCUT), Guangzhou, China

2020.09 - 2024.07

- Average grade ranking 3/66

Selected Publications

Beyond Mimicry: Learning Whole-Body Human-Humanoid Interaction from Human-Human Demonstrations

Wei-Jin Huang, Yue-Yi Zhang, Yi-Lin Wei, Zhi-Wei Xia, Juantao Tan, Yuan-Ming Li, Zhilin Zhao, Wei-Shi Zheng

CVPR 2026 - Proceedings of the Computer Vision and Pattern Recognition Conference

Abstract: Enabling humanoid robots to physically interact with humans is a critical frontier, but progress is hindered by the scarcity of high-quality human-humanoid interaction data. While leveraging abundant human-human interaction data presents a scalable alternative, we show that standard retargeting breaks essential contacts. We address this with PAIR and D-STAR for synchronized whole-body interaction behaviors.

Modeling Multiple Normal Action Representations for Error Detection in Procedural Tasks

Wei-Jin Huang, Yuan-Ming Li, Zhi-Wei Xia, Yu-Ming Tang, Kun-Yu Lin, Jian-Fang Hu, Wei-Shi Zheng

CVPR 2025 - Proceedings of the Computer Vision and Pattern Recognition Conference

Abstract: Error detection in procedural activities is essential for consistent and correct outcomes in AR-assisted and robotic systems. Existing methods often focus on temporal ordering errors or rely on static prototypes to represent normal actions. We propose an Adaptive Multiple Normal Action Representation framework that predicts all valid next actions and reconstructs their corresponding normal action representations for comparison with the ongoing action.

Less Static, More Private: Towards Transferable Privacy-Preserving Action Recognition by Generative Decoupled Learning

Zhi-Wei Xia, Kun-Yu Lin, Yuan-Ming Li, Wei-Jin Huang, Xian-Tuo Tan, Wei-Shi Zheng

ICCV 2025 - Proceedings of the IEEE/CVF International Conference on Computer Vision

Abstract: This work focuses on privacy-preserving action recognition, which aims to protect individual privacy in action videos without compromising recognition performance. Existing methods still struggle with video domain shifts. We propose GenPriv, a transferable framework that decouples static and dynamic video features and removes privacy-sensitive content from static action features.

Egoexo-fitness: Towards egocentric and exocentric full-body action understanding

Yuan-Ming Li, Wei-Jin Huang (Co-First), An-Lan Wang, Ling-An Zeng, Jing-Ke Meng, Wei-Shi Zheng

ECCV 2024 - European Conference on Computer Vision

Abstract: We present EgoExo-Fitness, a new full-body action understanding dataset featuring fitness videos recorded from synchronized egocentric and fixed exocentric cameras. Compared with existing datasets, EgoExo-Fitness not only contains first-person videos, but also provides rich annotations, including two-level temporal boundaries, natural language comments, and action quality scores.

Research Interests

Humanoid Learning: embodied intelligence, whole-body control, imitation learning, human-humanoid interaction; Video Understanding: action recognition, temporal action understanding, video representation learning

Experience

Algorithm Engineer Intern - China Telecom

2023.03 - 2023.05

Honors and Awards

2024

- President Scholarship, Sun Yat-sen University (top 5%)

2022

- Kaggle - H&M; Personalized Fashion Recommendations, Silver (85/2952)

- National Scholarship (1/66)

- Tencent Scholarship - First Class (1/66)

2021

- Kaggle - G2Net Gravitational Wave Detection, Silver (57/1219)